

**Module 7 Portfolio Project**

**School-Level Finance Data in Colorado Public Schools: Personnel Spending Patterns by School Locale**

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### **ABSTRACT**

This study analyzes Fiscal Year 2017 School-Level Finance Survey (SLFS) data to examine personnel spending patterns across Colorado public school locales. The analysis focuses on public, non-charter, regular (as opposed to alternative) schools and investigates whether per-pupil personnel expenditures, average teacher pay, and personnel spending allocations differ across urban, suburban, town, and rural schools. The study uses descriptive statistics, analysis of variance (ANOVA), Tukey's Honest Significant Difference (HSD) testing, and data visualizations generated in R Studio to identify and interpret spending patterns across locale categories.

Findings indicate statistically significant differences in per-pupil instructional, administrative, and student support expenditures across locales. Rural remote schools tended to exhibit higher per-pupil instructional spending, while large city schools showed higher administrative and student support expenditures. Results also revealed significant differences in average teacher pay across locales, with several rural locale categories exhibiting lower average teacher pay than urban and suburban schools. Additionally, locale was associated with differences in how schools allocate personnel spending across instructional, administrative, and student support functions.

These findings suggest that school locale is associated with differences in spending level, as well as with differences in personnel resource allocation priorities. Continued use of school-level finance data may support more transparent and informed decision-making related to staffing equity and school resource allocation.

## INTRODUCTION

School funding decisions influence the resources available to students, teachers, and school communities. Concerns about equity are especially important when comparing schools within and across districts. Schools may differ in staffing, instructional supports, technology access, and student services even when they serve similar student populations. Due to federal reporting requirements under the Every Student Succeeds Act (ESSA), schools are now required to share more detailed expenditure data than was previously reported (Cornman et al., 2021)

The School Level Finance Survey (SLFS) was developed by the National Center for Education Statistics (NCES) to provide school-level expenditure data that allows researchers and policymakers to examine how resources are distributed across schools (Cornman et al., 2021). The FY17 SLFS includes data on instructional salaries, teacher salaries, instructional aide salaries, student support services, school administration, technology, instructional materials and other non-personnel expenditures, as well as school context variables like student membership, teacher full-time equivalents, school level, locale, and charter status (Cornman et al., 2021). This creates an opportunity to explore variation in how much schools spend as well as what they prioritize. This topic is important because staffing structures and school resource allocation can shape teachers' working conditions and students' learning conditions.

Although school funding is discussed at the district or state level, those totals can hide important differences in how resources are distributed across individual schools. Two schools in the same district may operate under different conditions, creating a blind spot when only district level data is examined. The purpose of this project is to analyze the most recent available SLFS data, Fiscal Year 2017, to understand how school level financial decisions influence the staffing conditions that impact teaching and learning.

Findings around resource allocation patterns from this study can support leaders, organizations, and stakeholders in their conversations surrounding staffing equity, teacher support, strategic planning, and continuous improvement efforts. Should certain school contexts show lower instructional staff salaries or reduced investment in a particular personnel category, school leaders may need to revisit budgeting priorities. SLFS analysis can help accountability partners balance fiscal responsibility with equitable support for schools.

Community stakeholders including families, advocacy groups, and local policymakers are increasingly seeking transparency in school finance (Blad, 2025). This analysis will demystify the connections between locale, financial decisions, staffing conditions, administrative costs, and classroom support. This may help make school finance more understandable and actionable for those who are invested in the success of students and educators.

### **OBJECTIVES**

Prior analyses have focused on descriptive differences across school characteristics (Cornman, 2021). This study expands on that work by examining how per-pupil personnel expenditures, teacher compensation, and core personnel spending allocations differ across school locales in Colorado public schools. Specifically, the analysis investigates differences in per-pupil instructional, administrative, and student support expenditures, variation in average teacher pay across locales, and differences in how schools allocate personnel resources across instructional, administrative, and student support functions.

### **OVERVIEW OF STUDY (Due in Week Five)**

This study analyzes Fiscal Year 2017 SLFS data for Colorado public schools. The analysis focuses on public, non-charter regular schools, and uses school-level expenditure data to compare personnel spending patterns across urban, suburban, town, and rural school locales.

Because the SLFS reports expenditures at the individual school level, rather than broad district averages, it is possible to identify differences in spending patterns and resource allocation shares across locales that may be otherwise obscured.

The study investigates three related questions. First, it examines whether per-pupil personnel expenditures differ across locales. Second, it evaluates whether average teacher pay varies by locale. Third, it analyzes whether schools in different geographic contexts allocate different proportions of personnel spending toward instructional, administrative, and student support functions. Descriptive statistics, analysis of variance (ANOVA), Tukey comparison of means testing, and visualizations generated in R Studio are used to identify and interpret spending patterns across locale categories.

### **RESEARCH QUESTIONS AND HYPOTHESES**

- 1) Is school locale associated with variation in per-pupil instructional, student support, and administrative personnel expenditures?

H0<sub>1</sub>: There are no statistically significant differences in per-pupil instructional, student support, or administrative personnel expenditures associated with school locale.

H1<sub>1</sub>: There are statistically significant differences in per-pupil instructional, student support, or administrative personnel expenditures associated with school locale.

- 2) Is there an association between average teacher pay (pay\_per\_fte) and school locale?

H0<sub>2</sub>: There is no statistically significant association between average teacher pay and school locale.

H1<sub>2</sub>: There is a statistically significant association between average teacher pay and school locale.

- 3) Is there an association between school locale and the proportion of resources allocated to instructional, administrative, and student support functions?

H0<sub>3</sub>: There is no statistically significant association between school locale and the proportion of resources allocated to instructional, administrative, or student support functions.

H1<sub>3</sub>: There is a statistically significant association between school locale and the proportion of resources allocated to instructional, administrative, or student support functions.

## **LITERATURE REVIEW**

### **Research Question 1: Locale and Variation in School Expenditures**

Research suggests that school-level expenditures vary meaningfully based on locale, or the differences between urban, suburban, and rural schools. In her 2002 report on per-pupil spending in inner city and suburban schools, Marnie Shaul found that per-pupil expenditures were higher in the inner-city of four of the seven studied metropolitan areas as compared to the suburbs (Shaul, 2022). However, when weighted for student needs, the increased cost of special education and English Language Acquisition instruction, inner city schools were found to spend less overall.

More recent work using the SLFS reinforces the importance of examining expenditures by locale. The SLFS enables comparisons of expenditure variability across schools using locale, and school type (Cornman et al., 2021). This level of granularity allows researchers to compare similar schools, identifying patterns that would otherwise be obscured in district or state-level data.

Requirements of the Every Student Succeeds Act have created consistent reporting. This allows for reliable comparative analysis (Cornman et al., 2019). Cornman finds that most participating states have the capacity to report personnel and non-personnel expenditures across schools, particularly when allocation formulas are used instead of direct reporting. However, differences in attribution methods can introduce variation in reported expenditures, which should be considered when interpreting results (Atchison et al., 2017).

This literature supports the expectation that school locale is associated with differences in instructional, student support, and administrative expenditures. It also justifies the use of SLFS data and descriptive characteristics such as locale to analyze these differences.

**Research Question 2: Teacher Pay and Locale**

Research Question 2 examines whether teacher pay differs across school locales. Teacher salaries make up a substantial portion of school expenditures, and differences in compensation may reflect inequities in how resources are distributed across urban, suburban, town, and rural schools. Geographic context may influence salary patterns due to differences in labor markets, cost of living, district size and challenges related to teacher recruitment and retention.

Research on teacher wages highlights persistent disparities between teacher pay and that of other college-educated professionals. Research by Sylvia Allegretto found that teachers continue to experience a substantial wage penalty, earning only 73.4 cents for every dollar relative to the earnings of their peers (Allegretto, 2024). Lower pay may limit schools' ability to attract and retain teachers, particularly in geographically isolated or economically disadvantaged areas. This study examines whether average teacher pay differs systematically across Colorado school locales.

Long-term trends reinforce this pattern. Allegretto documents a steady increase in the teacher pay penalty over several decades, indicating that even with recent increases, compensation has not kept pace with other professions (Allegretto, 2024; Allegretto & Mishel, 2020). The existing research examines teacher pay at the state level, and does not test the impact of locale on teacher compensation.

**Research Question 3: Locale and Resource Allocation Shares**

The SLFS provides a valuable tool for analyzing allocation patterns, particularly the balance between instructional and administrative expenditures. According to Cornman, school-level finance data can be used to examine differences in how expenditures are distributed across categories such as instructional salaries, support services, and administrative costs (Cornman et al., 2021). The data enables comparisons of proportional spending that can reveal patterns in how schools allocate limited resources.

Policy-oriented research highlights the importance of allocation decisions. Rebecca Skinner and Wayne Riddle’s 2020 report on equitable per-pupil funding describes how weighted student funding systems are designed to allocate resources based on student needs, emphasizing instructional expenditures for disadvantaged populations (Skinner & Riddle, 2020). At the same time, studies of school finance during the Great Recession show that reductions in funding led to changes in spending priorities, with per-pupil spending declining by about \$600 per student in areas most impacted by the recession (Shores & Steinberg, 2019). These funding declines were associated with reductions in school resources and declines in student achievement, particularly in economically disadvantaged areas. The study does not isolate specific expenditure categories. However, the findings suggest that areas facing financial pressure were forced to make tradeoffs in how resources were allocated. This reinforces the importance of examining how schools balance instructional and administrative spending, with a focus on how locale may impact these trends.

## **RESEARCH DESIGN**

### **Methodology**

This study uses data from the School-Level Finance Survey (SLFS), which provides information on school-level expenditures, staffing, and enrollment. The dataset was filtered to include only public, non-charter schools in Colorado that were open at the beginning of the school year and classified as regular (not alternative) schools. Observations with zero enrollment or zero full-time equivalent teachers were removed to avoid division by zero errors and ensure meaningful per-pupil and ratio based calculations.

Key variables were constructed to support the research questions. Personnel expenditures were grouped into three categories: instructional (teacher salaries, instructional aides, instructional staff, and instructional staff support), administrative (principals, assistant principals, and secretaries), and student support (counselors, school psychologists, etc.). Core personnel spending was calculated as the sum of these categories. From these totals, per-pupil expenditures were derived by dividing each category by

student enrollment (MEMBER). Proportion variables were also calculated to measure the share of core personnel spending allocated to instructional, administrative, and student support functions.

Average teacher pay (pay\_per\_fte) was operationalized as teacher salary expenditures (Z39S) divided by full-time equivalent teachers (FTE). This variable was used to compare average teacher pay across school locales.

Locale classifications were based on the NCES Education Demographic and Geographic Estimates (EDGE) Program included in the SLFS dataset (Geverdt, 2018). Locale categories distinguish schools by geographic context and population characteristics. They include city, suburb, town, and rural designation. Subcategories identify relative size or geographic proximity. City and suburban municipalities can be large, midsize, or small. Towns and rural areas can be labeled as fringe, distant, or remote. These locale classifications were used as the primary grouping variable throughout the analysis to compare expenditure patterns, teacher pay, and personnel spending allocations across these geographic contexts.

## **Methods**

Three types of statistical analyses were conducted. First, descriptive statistics were generated to summarize central tendencies and variability across key variables, both overall and by school locale. Second, one-way analysis of variance (ANOVA) was used to evaluate whether per-pupil expenditures, average teacher pay, and spending proportions differed significantly across locale categories. When ANOVA results were significant, Tukey's Honest Significant Difference (HSD) test was applied to identify specific pairwise differences between locale groups. Tukey's HSD adjusts for family-wise error associated with multiple pairwise comparisons, reducing the likelihood of Type I error (Stoll, 2017) Analyses were conducted in R Studio using the psych, ggplot2 and dplyr packages.

The following derived variables will be created to standardize comparisons across schools (Fig. 1). Raw expenditure data can make direct comparisons misleading due to differences in school size.

**Figure 1***Derived Variables and source variables*

| Variable Name             | Description                                        | Formula                                                    |
|---------------------------|----------------------------------------------------|------------------------------------------------------------|
| instruction_per_pupil     | Instructional personnel spending                   | $(Z39 + Z40S + Z33S + V13S) / \text{member}$               |
| student_support_per_pupil | Student support personnel spending                 | $V11S / \text{member}$                                     |
| admin_per_pupil           | Administrative support service spending            | $V17S / \text{member}$                                     |
| pay_per_fte               | Average teacher salary                             | $Z39S / \text{FTE}$                                        |
| core_personnel            | Core personnel spending                            | $\text{instructional\_personnel spending} + V11S + V17S$   |
| instructional_prop        | Instructional proportion of core spending          | $\text{instructional\_personnel} / \text{core\_personnel}$ |
| admin_prop                | Administrative support proportion of core spending | $V17S / \text{core\_personnel}$                            |
| student_support_prop      | Student support proportion of core spending        | $V11S / \text{core\_personnel}$                            |

**Limitations**

The SLFS is incomplete, and findings should be interpreted accordingly without overstatement.

The data, though aggregated, represents real schools and communities. Discussions about findings should be constructive, and should avoid assigning blame to schools and educators. Although this is the most recent SLFS data available for download, it was collected during Fiscal Year 2017. It represents school finance patterns before the COVID-19 pandemic necessitated enormous investments in student technology, and a shift toward online curriculum. This study represents a snapshot of spending patterns that may not be reflective of current resource allocation priorities.

Additionally, locale classifications are broad categories that may not fully capture differences in community characteristics, regional labor markets, or district funding conditions across Colorado. Expenditure data also cannot directly measure educational quality, instructional effectiveness, or student outcomes. Because this analysis focuses only on Colorado public schools, findings may not generalize to other states with different funding systems or staffing structures.

### **Ethical Considerations**

Although the dataset does not contain any individuals' names or personal information, identities and/or salaries could be inferred if a specific school were to be discussed. For this reason, results will be discussed in terms of locale or district to ensure privacy.

Every effort will be made to avoid creating or adding to any stigmas associated with types of schools. This study does not seek to make claims that negatively influence funding decisions or public perception. Overclaiming must also be avoided. The project is observational, and the research is intended to reveal patterns without overstating causality.

### **FINDINGS**

Descriptive statistics indicate that instructional expenditures account for the majority of personnel spending across all schools, with proportions typically exceeding 85% (Fig. 2). Instructional spending per pupil were positively skewed (4.91) and high kurtosis (71.50), indicating a distribution with significant right-tail outliers. While the average instructional expenditure was approximately \$7,279 per pupil, the maximum exceeded \$40,000 per pupil, suggesting that a relatively small number of schools reported exceptionally high instructional expenditures. The median instructional expenditure (\$6,961) was also lower than the mean, further supporting the presence of positively skewed spending patterns.

Administrative spending per pupil also showed positive skewness (2.93), though the distribution was less extreme than instructional expenditures. The lower kurtosis value (15.43) suggests fewer severe outliers and more consistent administrative spending patterns across schools.

Student support expenditures were positively skewed (5.01) and high kurtosis (63.43), indicating that while many schools reported relatively low support-service expenditures, a small number of schools allocated substantially higher amounts toward student support personnel.

Average teacher pay per full-time equivalent teacher showed low, positive skew (0.26) and kurtosis of 5.11 indicating a more symmetric distribution across schools with outliers present. Teacher compensation patterns appeared more stable and normally distributed compared to per-pupil expenditure variables.

**Figure 2**

*Descriptive Statistics – Key Variables*

```
> # descriptive statistics for key variables
> describe(desc_vars)
```

|                           | vars | n    | mean     | sd       | median   | trimmed  | mad     | min    | max       | range     | skew  | kurtosis | se     |
|---------------------------|------|------|----------|----------|----------|----------|---------|--------|-----------|-----------|-------|----------|--------|
| instruction_per_pupil     | 1    | 1143 | 7278.50  | 1984.11  | 6961.11  | 7116.19  | 1381.30 | 420.42 | 40658.54  | 40238.12  | 4.91  | 71.50    | 58.69  |
| admin_per_pupil           | 2    | 1143 | 518.41   | 233.95   | 475.24   | 489.04   | 143.60  | 0.00   | 2390.24   | 2390.24   | 2.93  | 15.43    | 6.92   |
| student_support_per_pupil | 3    | 1143 | 228.56   | 218.40   | 188.68   | 199.11   | 154.56  | 0.00   | 3779.07   | 3779.07   | 5.01  | 63.42    | 6.46   |
| pay_per_fte               | 4    | 1143 | 55946.62 | 11207.30 | 55067.11 | 55277.31 | 8118.77 | 0.00   | 132347.83 | 132347.83 | 0.26  | 5.11     | 331.50 |
| instructional_prop        | 5    | 1143 | 0.91     | 0.04     | 0.91     | 0.91     | 0.03    | 0.40   | 1.00      | 0.60      | -3.94 | 37.11    | 0.00   |
| admin_prop                | 6    | 1143 | 0.07     | 0.03     | 0.06     | 0.06     | 0.02    | 0.00   | 0.36      | 0.36      | 2.84  | 21.19    | 0.00   |
| student_support_prop      | 7    | 1143 | 0.03     | 0.02     | 0.02     | 0.02     | 0.02    | 0.00   | 0.36      | 0.36      | 4.04  | 40.29    | 0.00   |

```
> |
```

Grouped descriptive statistics in the R Studio output show that rural and town schools tend to allocate a higher proportion of personnel spending to instruction, while large city schools allocate more to administrative and student support personnel spending (Fig. 3). Rural remote schools reported the highest average instructional spending per pupil (\$10,365) as well as the greatest variability in instructional expenditures (sd = 7,876) (Fig. 3). The large standard deviation points to highly uneven spending patterns among rural remote schools. Rural distant schools also demonstrated comparatively high instructional expenditures (\$8,049 per pupil, sd=2928). In contrast, suburban midsize and town fringe schools reported lower variability in instructional spending (sd = 918 and sd = 850), suggesting more consistent spending patterns across schools within those locale categories.

Administrative standard deviations are comparatively smaller and stable. For example, town fringe schools reported an administrative spending standard deviation of approximately 123, while rural remote schools showed the greatest administrative variability with a standard deviation of 599 (Fig. 3).

This may be due to more consistent administrative staffing structures across schools regardless of locale.

Student support expenditures were highest among large city schools, which reported average student support spending of approximately \$344 per pupil (Fig. 3). By comparison, rural remote schools reported average student support expenditures of approximately \$163 per pupil, while fringe town schools averaged approximately \$144 per pupil. These differences suggest variation in access to counselors, psychologists, and other student support personnel across geographic contexts. Boxplots further illustrate differences in per-pupil spending across locales, showing wider variation in instructional expenditures compared to administrative spending (Figs. 4, 5, & 6).

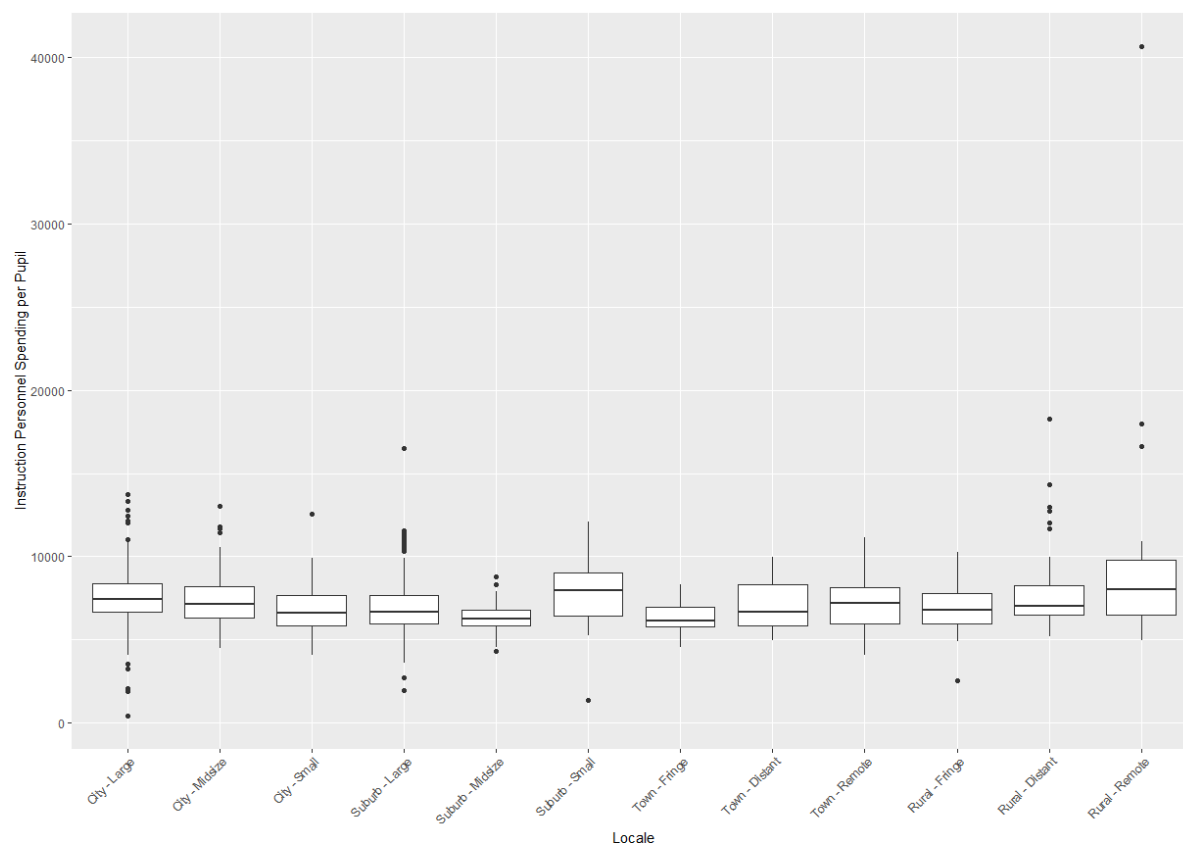
**Figure 3**

*Grouped Descriptive Statistics – Means and Standard Deviations by Locale*

|    | LOCALE         | mean_instruction | sd_instruction | mean_admin | sd_admin | mean_support | sd_support |
|----|----------------|------------------|----------------|------------|----------|--------------|------------|
|    | <fct>          | <dbl>            | <dbl>          | <dbl>      | <dbl>    | <dbl>        | <dbl>      |
| 1  | City - Large   | 7582.            | 1604.          | 601.       | 261.     | 344.         | 234.       |
| 2  | City - Mids... | 7487.            | 1515.          | 509.       | 161.     | 180.         | 141.       |
| 3  | City - Small   | 6957.            | 1769.          | 446.       | 130.     | 166.         | 107.       |
| 4  | Suburb - La... | 6977.            | 1651.          | 485.       | 199.     | 191.         | 173.       |
| 5  | Suburb - Mi... | 6355.            | 918.           | 447.       | 286.     | 241.         | 581.       |
| 6  | Suburb - Sm... | 7817.            | 2016.          | 503.       | 155.     | 200.         | 146.       |
| 7  | Town - Frin... | 6325.            | 850.           | 431.       | 123.     | 144.         | 96.6       |
| 8  | Town - Dist... | 7058.            | 1531.          | 386.       | 174.     | 162.         | 187.       |
| 9  | Town - Remo... | 7129.            | 1601.          | 475.       | 159.     | 176.         | 111.       |
| 10 | Rural - Fri... | 6921.            | 1299.          | 499.       | 164.     | 193.         | 128.       |
| 11 | Rural - Dis... | 8049.            | 2928.          | 550.       | 372.     | 184.         | 137.       |
| 12 | Rural - Rem... | 10365.           | 7876.          | 631.       | 599.     | 163.         | 259.       |

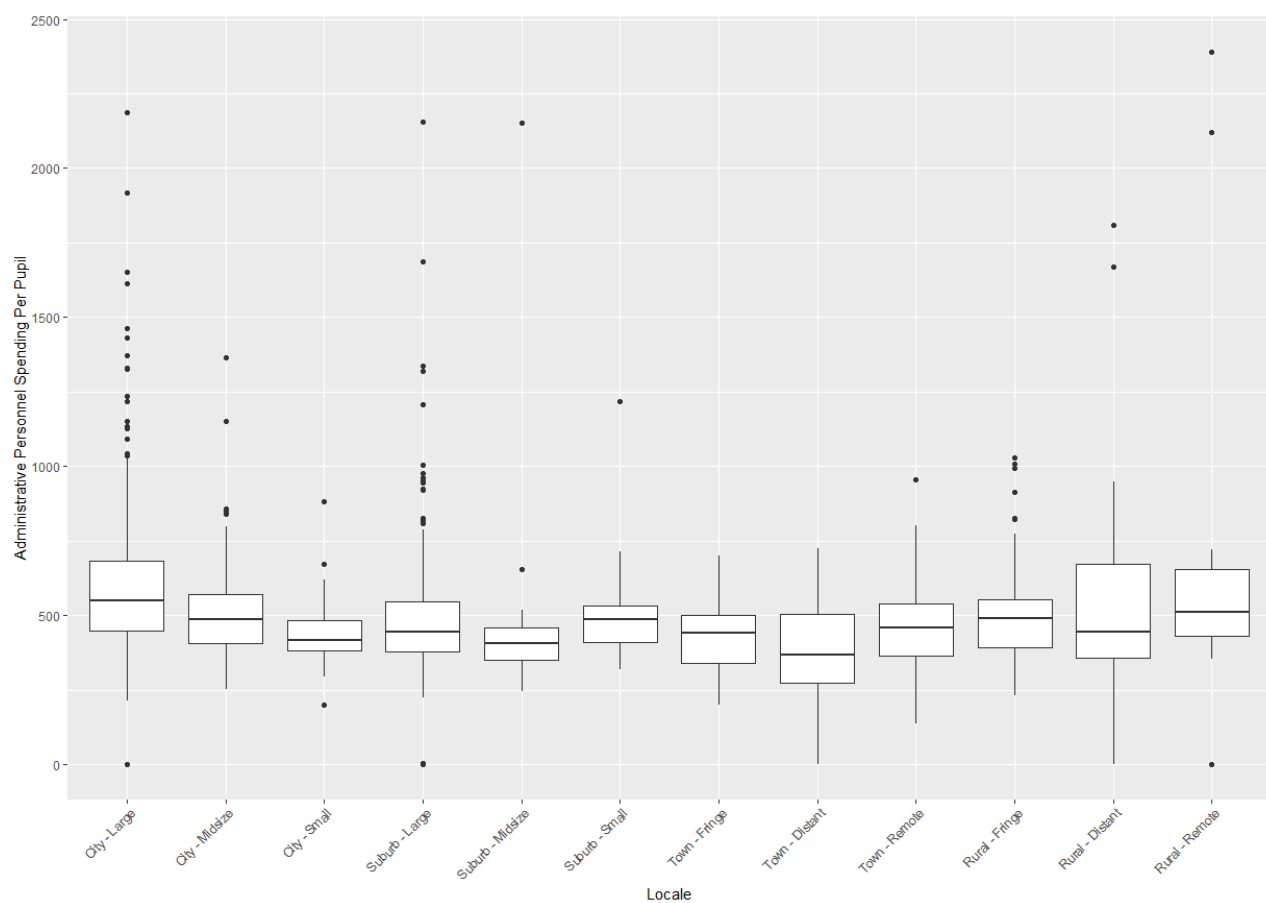
**Figure 4**

*Boxplot – Distribution of Instruction Personnel Spending Per Pupil by Locale*



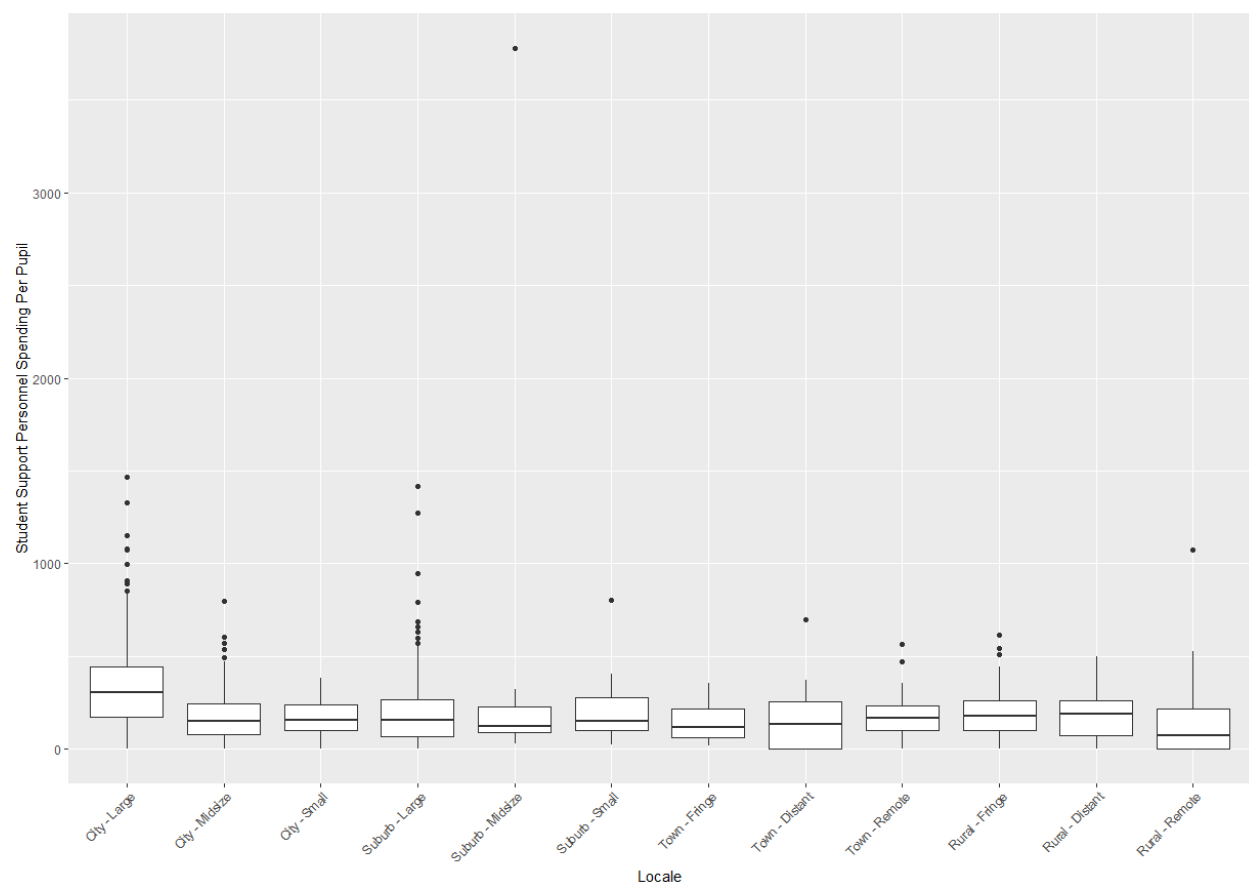
**Figure 5**

*Boxplot – Administrative Personnel Spending Per Pupil by Locale*



**Figure 6**

*Boxplot – Student Support Personnel Spending Per Pupil by Locale*



### Research Question 1: Is school locale associated with variation in per-pupil expenditures?

The ANOVA results indicate statistically significant differences in per-pupil instructional ( $F = 9.11$ ,  $p < 0.001$ ), administrative ( $F = 6.7$ ,  $p < 0.001$ ), and student support ( $F = 12.02$ ,  $p < 0.001$ ) expenditures across school locales (Fig. 7). The null hypothesis can be rejected. Among these categories, student support expenditures demonstrated the strongest overall locale effect, indicated by the largest ANOVA F-statistic.

**Figure 7**

*ANOVA Output*

```
> anova_instr <- aov(instruction_per_pupil ~ LOCALE, data = colosLFSreduced)
> summary(anova_instr)
              Df      Sum Sq Mean Sq F value    Pr(>F)
LOCALE         11  3.661e+08  33279573    9.114 9.03e-16 ***
Residuals    1131  4.130e+09   3651316
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
> anova_admin <- aov(admin_per_pupil ~ LOCALE, data = colosLFSreduced)
> summary(anova_admin)
              Df      Sum Sq Mean Sq F value    Pr(>F)
LOCALE         11   3823824   347620    6.7 5.84e-11 ***
Residuals    1131  58678509    51882
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
> anova_support <- aov(student_support_per_pupil ~ LOCALE, data = colosLFSreduced)
> summary(anova_support)
              Df      Sum Sq Mean Sq F value    Pr(>F)
LOCALE         11  5699010   518092   12.02 <2e-16 ***
Residuals    1131  48770861    43122
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Further analysis using Tukey multiple comparison of means testing reveals that instructional spending differences are most pronounced in rural remote schools. These schools reported significantly higher average instructional expenditures than large city schools, with mean differences exceeding \$2700 per pupil in several comparisons (Fig. 8).

**Figure 8**

*Tukey Multiple Means Comparison – Instruction Per Pupil – Sorted by p-value*

|                                  | diff        | lwr         | upr          | p adj        |
|----------------------------------|-------------|-------------|--------------|--------------|
| Rural - Remote-Suburb - Large    | 3388.61508  | 1950.76227  | 4826.4678907 | 1.953659e-12 |
| Rural - Remote-Suburb - Midsize  | 4010.05114  | 2296.26876  | 5723.8335213 | 2.807643e-12 |
| Rural - Remote-Town - Fringe     | 4040.19787  | 2233.71263  | 5846.6831140 | 3.025680e-11 |
| Rural - Remote-Rural - Fringe    | 3444.56275  | 1896.00185  | 4993.1236475 | 4.004641e-11 |
| Rural - Remote-Town - Remote     | 3236.58178  | 1610.21693  | 4862.9466242 | 7.087422e-09 |
| Rural - Remote-City - Large      | 2783.44131  | 1338.25312  | 4228.6295045 | 2.672761e-08 |
| Rural - Remote-City - Midsize    | 2877.92519  | 1374.06203  | 4381.7883614 | 3.434475e-08 |
| Rural - Remote-City - Small      | 3408.32477  | 1576.21434  | 5240.4351971 | 1.000783e-07 |
| Rural - Remote-Town - Distant    | 3307.26949  | 1302.49492  | 5312.0440706 | 5.204399e-06 |
| Rural - Remote-Suburb - Small    | 2548.37581  | 819.62561   | 4277.1260004 | 9.967190e-05 |
| Rural - Remote-Rural - Distant   | 2316.04553  | 561.93334   | 4070.1577178 | 1.008455e-03 |
| Suburb - Large-City - Large      | -605.17377  | -1094.99302 | -115.3545241 | 3.220548e-03 |
| Rural - Distant-Suburb - Midsize | 1694.00561  | 245.59514   | 3142.4160812 | 7.473581e-03 |
| Suburb - Midsize-City - Large    | -1226.60983 | -2279.96269 | -173.2569779 | 7.956190e-03 |
| Rural - Distant-Town - Fringe    | 1724.15234  | 167.15787   | 3281.1468103 | 1.571032e-02 |
| Town - Fringe-City - Large       | -1256.75656 | -2455.04331 | -58.4698128  | 3.014752e-02 |
| Suburb - Small-Suburb - Midsize  | 1461.67534  | 44.08564    | 2879.2650354 | 3.636357e-02 |
| Suburb - Midsize-City - Midsize  | -1132.12595 | -2264.63985 | 0.3879562    | 5.017172e-02 |

Note: Tukey testing identified 18 statistically significant relationships out of 66 comparisons

Administrative expenditure differences were also statistically significant across locales, though, the size of the differences were smaller than those observed for instructional spending (Figs. 7 & 9). Tukey comparisons show that large city schools report significantly higher administrative expenditures per pupil than many suburban, town and rural locales (Fig. 9).

**Figure 9**

*Tukey Multiple Means Comparison – Admin Per Pupil – Sorted by p-value*

|                               | diff        | lwr         | upr        | p adj        |
|-------------------------------|-------------|-------------|------------|--------------|
| Suburb - Large-City - Large   | -116.433775 | -174.821208 | -58.046342 | 6.503410e-09 |
| Suburb - Midsize-City - Large | -154.098817 | -279.660582 | -28.537052 | 3.587884e-03 |
| Town - Distant-City - Large   | -215.609845 | -392.078059 | -39.141632 | 3.853145e-03 |
| Town - Fringe-City - Large    | -170.385003 | -313.223177 | -27.546828 | 5.586027e-03 |
| City - Midsize-City - Large   | -92.368145  | -170.906489 | -13.829801 | 6.892836e-03 |
| Town - Remote-City - Large    | -125.723704 | -233.505305 | -17.942103 | 7.764255e-03 |
| Rural - Fringe-City - Large   | -102.004310 | -192.042705 | -11.965914 | 1.165174e-02 |
| City - Small-City - Large     | -155.513089 | -302.915931 | -8.110248  | 2.819312e-02 |
| Rural - Remote-Town - Distant | 245.134364  | 6.161229    | 484.107500 | 3.843676e-02 |

Note: Tukey testing identified nine statistically significant relationships out of 66 comparisons.

Student support expenditures demonstrated the strongest locale effect based on the ANOVA F-statistic of 12.02 ( $p < 0.001$ ). Tukey comparisons indicate that large city schools reported significantly higher students support expenditures than many suburban, town, and rural locale categories (Fig. 10). Large city schools spent \$181 more per pupil on students support services than rural remote schools and approximately \$200 more per pupil than town fringe schools. The consistency of these significant comparisons across multiple locale groups points to a systematic difference between city schools and other locales.

**Figure 10**

*Tukey Multiple Means Comparison – Student Support Per Pupil – Sorted by p-value*

|                              | diff        | lwr        | upr        | p adj        |
|------------------------------|-------------|------------|------------|--------------|
| Suburb - Large-City - Large  | -153.252696 | -206.48314 | -100.02226 | 2.610134e-13 |
| City - Midsize-City - Large  | -164.050316 | -235.65186 | -92.44877  | 8.512191e-12 |
| Rural - Fringe-City - Large  | -151.038518 | -233.12439 | -68.95265  | 1.493137e-07 |
| Town - Remote-City - Large   | -168.451441 | -266.71337 | -70.18951  | 1.622357e-06 |
| Town - Fringe-City - Large   | -200.485326 | -330.70751 | -70.26314  | 3.442648e-05 |
| City - Small-City - Large    | -178.791747 | -313.17543 | -44.40807  | 8.732670e-04 |
| Rural - Distant-City - Large | -160.659416 | -282.13165 | -39.18719  | 9.767487e-04 |
| Suburb - Small-City - Large  | -144.803366 | -261.90290 | -27.70383  | 3.175051e-03 |
| Rural - Remote-City - Large  | -181.015845 | -338.06971 | -23.96198  | 9.201227e-03 |
| Town - Distant-City - Large  | -182.631940 | -343.51383 | -21.75005  | 1.133819e-02 |

Note: Tukey testing identified ten statistically significant relationships out of 66 comparisons.

### **Research Question 2: Is there an association between average teacher pay and school locale?**

ANOVA results indicate statistically significant differences in average teacher pay across school locales ( $p < 0.001$ ) (Fig. 11). The null hypothesis can be rejected. Tukey multiple comparison of means identified nine statistically significant pairwise differences out of 66 comparisons (Fig. 12) Figure 12 shows that average teacher pay differs most substantially between rural/town and urban/suburban locales, with several rural locale categories exhibiting lower average teacher pay than large city and suburban schools.

These findings suggest that geographic context may influence teacher compensation patterns in Colorado public schools. Differences in average teacher pay may reflect variation in local labor markets, district resources, recruitment conditions, and regional cost-of-living adjustments. Teacher education level, years of experience, and district salary schedules also contribute to teacher pay and may contribute to variation in compensation across locales.

**Figure 11**

*ANOVA Results – Pay per FTE by LOCALE*

```
> summary(anova_pay_per_FTE)
              Df      Sum Sq   Mean Sq F value    Pr(>F)
LOCALE         11  8.434e+09  766769240    6.424 2.05e-10 ***
Residuals     1131 1.350e+11 119367717
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

**Figure 12**

*Tukey Comparison of Means – Pay per FTE ~ Locale – Sorted by p-value*

|                                 | diff          | lwr          | upr        | p adj        |
|---------------------------------|---------------|--------------|------------|--------------|
| Town - Remote-Suburb - Small    | -13727.453408 | -21220.80939 | -6234.0974 | 1.747865e-07 |
| Town - Remote-Suburb - Large    | -8871.776782  | -13974.34589 | -3769.2077 | 1.033879e-06 |
| Town - Remote-City - Large      | -8593.754913  | -13763.62751 | -3423.8823 | 4.161384e-06 |
| Town - Remote-City - Midsize    | -9194.888859  | -14885.61882 | -3504.1589 | 9.438492e-06 |
| Suburb - Small-Suburb - Midsize | 10598.943098  | 2493.63933   | 18704.2469 | 1.206335e-03 |
| Town - Distant-Suburb - Small   | -12895.281029 | -22948.67304 | -2841.8890 | 1.706628e-03 |
| Rural - Fringe-Suburb - Small   | -7752.385519  | -14685.98653 | -818.7845  | 1.387117e-02 |
| Rural - Distant-Suburb - Small  | -9085.816869  | -17468.42951 | -703.2042  | 2.051640e-02 |
| Town - Fringe-Suburb - Small    | -8962.460163  | -17701.13981 | -223.7805  | 3.850632e-02 |

Note: Tukey testing identified nine statistically significant relationships out of 66 comparisons.

### **Research Question 3: Is there an association between school locale and the proportion of resources allocated across personnel categories?**

ANOVA results indicate statistically significant differences in how schools allocate personnel spending across instructional ( $F = 8.52$ ,  $p < 0.001$ ), administrative ( $F = 2.727$ ,  $p = 0.002$ ), and student support groups ( $F = 10.8$ ,  $p < 0.001$ )(Fig. 13). The null hypothesis is rejected.

**Figure 13**

*ANOVA Output – Instructional Proportion of Personnel Spending by Locale*

```
> anova_instr_prop <- aov(instructional_prop ~ LOCALE, data = colosLFSreduced)
> summary(anova_instr_prop)
              Df Sum Sq Mean Sq F value    Pr(>F)
LOCALE         11  0.1468  0.013344    8.523 1.38e-14 ***
Residuals    1131  1.7708  0.001566
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

> anova_admin_prop <- aov(admin_prop ~ LOCALE, data = colosLFSreduced)
> summary(anova_admin_prop)
              Df Sum Sq Mean Sq F value    Pr(>F)
LOCALE         11  0.0205  0.0018631    2.727 0.00175 **
Residuals    1131  0.7726  0.0006831
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

> anova_support_prop <- aov(student_support_prop ~ LOCALE, data = colosLFSreduced)
> summary(anova_support_prop)
              Df Sum Sq Mean Sq F value    Pr(>F)
LOCALE         11  0.0626  0.005694    10.8 <2e-16 ***
Residuals    1131  0.5962  0.000527
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Descriptive statistics show that instructional spending accounted for the largest share of personnel expenditures across all locales, ranging from 89.9% of core personnel spending in large city schools to 93.1% in distant town schools (Fig. 14)

**Figure 14**

*Distribution of Proportional Variables by Locale*

| LOCALE<br><fct>    | mean_instr_prop<br><dbl> | sd_instr_prop<br><dbl> | mean_admin_prop<br><dbl> | sd_admin_prop<br><dbl> | mean_student_support_prop<br><dbl> | sd_student_support_prop<br><dbl> |
|--------------------|--------------------------|------------------------|--------------------------|------------------------|------------------------------------|----------------------------------|
| 1 City - Large     | 0.889                    | 0.0423                 | 0.0718                   | 0.0306                 | 0.0395                             | 0.0241                           |
| 2 City - Midsize   | 0.916                    | 0.0250                 | 0.0627                   | 0.0168                 | 0.0217                             | 0.0153                           |
| 3 City - Small     | 0.918                    | 0.0256                 | 0.0604                   | 0.0186                 | 0.0213                             | 0.0134                           |
| 4 Suburb - Large   | 0.911                    | 0.0341                 | 0.0651                   | 0.0246                 | 0.0240                             | 0.0209                           |
| 5 Suburb - Midsize | 0.907                    | 0.0802                 | 0.0625                   | 0.0272                 | 0.0306                             | 0.0553                           |
| 6 Suburb - Small   | 0.908                    | 0.0866                 | 0.0648                   | 0.0502                 | 0.0274                             | 0.0371                           |
| 7 Town - Fringe    | 0.917                    | 0.0241                 | 0.0624                   | 0.0163                 | 0.0206                             | 0.0138                           |
| 8 Town - Distant   | 0.931                    | 0.0263                 | 0.0502                   | 0.0193                 | 0.0187                             | 0.0206                           |
| 9 Town - Remote    | 0.916                    | 0.0246                 | 0.0615                   | 0.0173                 | 0.0227                             | 0.0139                           |
| 10 Rural - Fringe  | 0.909                    | 0.0278                 | 0.0662                   | 0.0191                 | 0.0251                             | 0.0161                           |
| 11 Rural - Distant | 0.914                    | 0.0366                 | 0.0632                   | 0.0284                 | 0.0224                             | 0.0166                           |
| 12 Rural - Remote  | 0.927                    | 0.0320                 | 0.0592                   | 0.0306                 | 0.0133                             | 0.0163                           |

Tukey multiple means comparisons show that all the significant instructional proportion differences involve large city schools, with large cities allocating a smaller share to instruction than all

town and rural locales, and also less than large suburb schools, midsize city schools, and small city schools (Fig. 15). These comparative differences show that rural and remote schools allocate larger shares of personnel expenditures directly toward instructional personnel.

**Figure 15**

*Tukey Multiple Means Comparison – Instruction Proportion – Sorted by p-value*

|                              | diff         | lwr         | upr         | p adj        |
|------------------------------|--------------|-------------|-------------|--------------|
| Suburb - Large-City - Large  | 0.0221368762 | 0.011993966 | 0.032279786 | 1.045599e-10 |
| City - Midsize-City - Large  | 0.0268354309 | 0.013191958 | 0.040478904 | 1.150482e-08 |
| Town - Remote-City - Large   | 0.0270084786 | 0.008284944 | 0.045732013 | 1.638905e-04 |
| Town - Distant-City - Large  | 0.0423104971 | 0.011654906 | 0.072966088 | 4.227967e-04 |
| Rural - Remote-City - Large  | 0.0385160474 | 0.008589877 | 0.068442218 | 1.607274e-03 |
| Rural - Fringe-City - Large  | 0.0199517621 | 0.004310531 | 0.035592994 | 1.880786e-03 |
| City - Small-City - Large    | 0.0293251978 | 0.003718767 | 0.054931629 | 1.005499e-02 |
| Town - Fringe-City - Large   | 0.0281845161 | 0.003371047 | 0.052997986 | 1.124905e-02 |
| Rural - Distant-City - Large | 0.0257347840 | 0.002588593 | 0.048880975 | 1.491777e-02 |

Note: Tukey testing identified nine statistically significant relationships out of 66 comparisons

Administrative spending proportions showed weaker overall locale effects than instructional or student support proportions. The ANOVA was significant ( $p = 0.002$ ), and Tukey testing identified two significant differences involving large city schools (Fig. 16). Large city schools spent a larger share of core personnel expenditures when compared to schools in distant towns ( $p = 0.029$ ) and midsize cities ( $p = 0.048$ )(Fig. 16) Large city schools reported the highest average administrative spending proportion (0.072), compared to town distant schools who reported the lowest administrative proportion (0.050)(Fig. 14). The smaller differences and fewer significant Tukey comparisons suggest that spending administrative spending allocations are more stable across locales than instructional and student support expenditure shares.

**Figure 16**

*Tukey Multiple Means Comparison – Administrative Proportion – Sorted by p-value*

|                             | diff          | lwr          | upr           | p adj      |
|-----------------------------|---------------|--------------|---------------|------------|
| Town - Distant-City - Large | -2.154120e-02 | -0.041790006 | -1.292388e-03 | 0.02561989 |
| City - Midsize-City - Large | -9.046549e-03 | -0.018058415 | -3.468245e-05 | 0.04810495 |

Note: Tukey testing identified two statistically significant relationships out of 66 comparisons

Students support spending proportions showed clear locale-based differences identified through Tukey testing (Fig. 17). Large city schools reported significantly higher student support spending proportions than large suburbs ( $p < 0.001$ ), small and midsize cities ( $p = 0.005$ ,  $p < 0.001$ ), all town locales, fringe, distant, and remote ( $p < 0.001$ ,  $p = 0.008$ ,  $p < 0.001$ ), and all rural locales, fringe, distant, and remote ( $p < 0.001$ ,  $p = 0.002$ ,  $p < 0.001$ )(Fig.17). Large cities allocated 3.95% of core personnel spending to student support staff salaries, compared to the 1.35% allocated to student support functions in rural remote schools (Fig. 14). This suggests that student support allocations vary systematically across locales.

**Figure 17**

*Tukey Multiple Means Comparison – Student Support Proportion – Sorted by p-value*

|                              | diff          | lwr          | upr           | p adj        |
|------------------------------|---------------|--------------|---------------|--------------|
| Suburb - Large-City - Large  | -0.0154724404 | -0.021357862 | -0.0095870185 | 2.565725e-13 |
| City - Midsize-City - Large  | -0.0177888823 | -0.025705505 | -0.0098722595 | 2.370670e-11 |
| Rural - Fringe-City - Large  | -0.0143596843 | -0.023435506 | -0.0052838625 | 1.681582e-05 |
| Town - Remote-City - Large   | -0.0167526578 | -0.027616985 | -0.0058883305 | 3.306873e-05 |
| Rural - Remote-City - Large  | -0.0259750422 | -0.043339697 | -0.0086103874 | 7.036959e-05 |
| Town - Fringe-City - Large   | -0.0188531257 | -0.033251137 | -0.0044551145 | 1.176703e-03 |
| Rural - Distant-City - Large | -0.0171249525 | -0.030555526 | -0.0036943795 | 1.894068e-03 |
| City - Small-City - Large    | -0.0179565649 | -0.032814692 | -0.0030984378 | 4.574123e-03 |
| Town - Distant-City - Large  | -0.0207693005 | -0.038557201 | -0.0029813994 | 7.655894e-03 |

Note: Tukey testing identified nine statistically significant relationships out of 66 comparisons

## CONCLUSION

Findings from research question one indicate that per-pupil spending differs significantly across school locales. Instructional spending was highest in rural remote schools, while administrative and student support personnel expenditures were higher in large city schools. These results suggest that locale is associated with differences in how much schools spend per student across key personnel categories.

Research question two identified statistically significant differences in average teacher pay across locales. This finding suggests that teacher compensation varies geographically, with some locale categories exhibiting lower average teacher pay than others. These differences may reflect broader disparities in labor markets, recruitment challenges, and district resources.

Research question three further shows that differences across locales extend to spending proportions. Rural schools allocate a greater proportion of personnel spending to instructional staff, while large city schools allocate relatively more to administrative and student support service staffing. These findings suggest that schools in different contexts are making tradeoffs in how resources are distributed.

In all, the findings from this study reveal that locale is associated with differences in per-pupil expenditures, teacher compensation, and spending priorities. Rural remote schools demonstrated particularly high instructional spending variability, while large city schools allocated greater shares of personnel spending toward student support functions. Administrative spending was more stable, with large city schools spending more than other groups.

The use of school-level finance data provides a more detailed understanding of expenditure patterns than district-level summaries alone. By examining individual schools rather than district averages, the analysis can detect variation that may be hidden within broader financial reporting systems. These findings reaffirm the importance of transparent school-finance reporting for evaluating resource allocation priorities and educational equity across Colorado public schools.

## RECOMMENDATIONS

Based on these findings, schools and districts should continue to prioritize financial reporting. The SLFS provides a valuable tool for understanding how resources are distributed. Continued use of school-level data can support more informed decision making. Making patterns visible and accessible to stakeholders, including families and community members can strengthen accountability and trust.

District leaders and policymakers should also examine whether teacher compensation patterns differ systematically across geographic contexts. Persistent differences in teacher pay may be associated with recruitment and retention challenges, particularly in rural, remote schools. Additional analysis may help determine whether compensation disparities are in line with differences in cost-of-living, and if they are associated with staffing shortages or differences in educator experience.

Additional investigation is recommended surrounding the cause of the disparity between rural/town and urban/suburban spending on support services. More specifically, future research should seek to understand if students support services in rural areas are not needed because of demographic differences, or if rural schools are unable to recruit and retain students support service providers. Future research should build on this study's findings by incorporating additional variables such as student demographics, achievement outcomes, and post-pandemic spending patterns. Because this study uses fiscal year 2017 data, the most recent available for public download, further analysis incorporating longitudinal data is needed to determine whether these patterns have shifted in response to recent changes in school funding and instructional delivery.

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